A PROJECT TITLE
ON
FERTILIZER RECOMMENDATION SYSTEM USING MACHINE LEARNING

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY
In
A.Y.: 2026-27

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Abstract

This project builds a multi-class classification system that recommends a fertilizer from soil, crop and nutrient measurements, and Random Forest was the most accurate of three classifiers at 94.5% test accuracy. Selecting the right fertilizer for a given crop and soil condition is a critical yet often inefficient process for farmers, who traditionally rely on generic guidelines or trial-and-error methods that can lead to poor yield, wasted resources, and soil degradation. The system predicts the most suitable fertilizer from eight inputs: soil type, crop type, temperature, humidity, moisture, and nitrogen, phosphorus and potassium levels.

The dataset has 1,000 records and seven fertilizer classes (10-26-26, 14-35-14, 17-17-17, 20-20, 28-28, DAP and Urea). It is a rule-based synthetic dataset with 5% label noise, so the pipeline runs without an external file. The data is cleaned and inspected, categorical features are label-encoded, numerical features are standardised, and an 80:20 stratified split is used. Logistic Regression, Decision Tree and Random Forest are trained and compared on accuracy, precision, recall and F1-score. Random Forest scored 94.5% accuracy and 0.941 weighted F1, ahead of the Decision Tree (91.0%) and Logistic Regression (71.0%). The best model powers a simple recommendation function that returns a fertilizer suggestion from user-supplied soil and crop details.

By automating this decision, the system aims to help farmers make informed, data-driven choices, improve crop productivity, and promote sustainable and cost-effective use of fertilizers.

Keywords**:** Fertilizer Recommendation; Machine Learning; Random Forest; Decision Tree; Logistic Regression; Multi-class Classification; Precision Agriculture

1.Introduction

1.1 Purpose of the Project

The purpose is to recommend a suitable fertilizer from several soil, crop and weather measurements at once, because no single reading is enough to choose correctly. The system uses eight inputs: temperature, humidity, moisture, soil type, crop type, nitrogen, potassium and phosphorous. From these it learns which of seven fertilizers (10-26-26, 14-35-14, 17-17-17, 20-20, 28-28, DAP, Urea) fits a given condition.

The motivation is to replace generic guidelines and trial-and-error with a quick, data-driven suggestion that supports farmers in saving input cost and protecting soil health. The output is a decision aid, not a substitute for a certified soil-testing laboratory or an agronomist.

1.2 Problem Statement

The problem is a multi-class supervised classification task. Given the temperature, humidity, moisture, soil type, crop type, and nitrogen, potassium and phosphorous levels of a field, the system predicts the fertilizer name (one of seven classes) that best suits that condition. Three classifiers are trained and compared, and the best one is used for the recommendation function.

1.3 Existing Methods and Disadvantages

Farmers commonly choose fertilizers using generic regional guidelines, past experience, shopkeeper advice or trial-and-error. Laboratory soil testing gives more reliable nutrient values, but turning those values into a fertilizer choice for a particular crop and weather condition is still done by hand.

These approaches have clear disadvantages: recommendations are not tailored to the specific soil-crop-nutrient combination, trial-and-error can waste fertilizer and reduce yield, and over- or under-application can degrade the soil. This project does not include a separate conventional baseline experiment, so no numerical performance disadvantage is claimed.

Machine learning can learn these relationships directly from data and give an instant, consistent recommendation. This project does not include a separate conventional baseline experiment, so no numerical performance disadvantage of the traditional methods is claimed.

1.4 Proposed System

The proposed system is a simple, reproducible machine learning pipeline that turns soil, crop and weather readings into a fertilizer recommendation. It works in the following stages:

1. **Dataset:** a dataset of 1,000 records with eight input features and seven fertilizer classes is built, with a fixed random seed so the results can be repeated.
2. **Inspection and EDA**: the data is inspected for shape, data types and missing values, and explored through the class distribution and the correlation between numerical features.
3. **Preprocessing**: the categorical features (soil type, crop type and the fertilizer name) are label-encoded, and the numerical features are standardised so all inputs are on a comparable scale.
4. **Splitting**: the data is divided 80:20 into training and test sets with stratification, so every fertilizer class appears in both sets in the same proportion.
5. **Training:** Logistic Regression, Decision Tree and Random Forest classifiers are trained on the same training data, covering a linear model, a single tree and an ensemble of trees.
6. **Evaluation and selection:** each model is tested on the held-out data with accuracy, precision, recall and F1-score, and the most accurate model is selected automatically.
7. **Analysis:** the best model is examined with a confusion matrix, a per-class report and a feature-importance chart to show where it succeeds and which inputs matter most.
8. **Recommendation:** the best model is wrapped in a function that accepts raw soil and crop details, applies the same encoding and scaling, and returns the recommended fertilizer name instantly.

1.5 Research Objectives(atleast 4)

1. Prepare and inspect the fertilizer dataset and identify the target variable (Fertilizer_Name).
2. Verify data quality (missing values, data types, statistical summary) and explore class distribution and feature correlations.
3. Encode categorical features and scale numerical features so the models receive well-structured input.
4. Prepare a stratified 80:20 train-test split.
5. Train Logistic Regression, Decision Tree and Random Forest classifiers on the prepared data.
6. Evaluate and compare the models using accuracy, precision, recall and F1-score, and identify the best model.
7. Analyse the best model with a confusion matrix and feature importance, and build a recommendation function for new soil and crop inputs.

2. Literature Survey (recent studies from 2-3 years)

Ten recent studies (2024–2026) show that tree-based ensembles and, increasingly, explainable deep models give the best fertilizer and crop recommendations, which supports comparing Random Forest and Decision Tree with a linear baseline in this project. Where a paper's own future-work statement was not available, the last column gives a suggested direction, marked (suggested).

S.No.	Title	Reference with DOI	Problem or gap addressed	Objective	Methods used	Inference	Recommendations for future research
1	Interpretable deep learning models for independent fertilizer and crop recommendation	Venkateswara, S. M., & Padmanaban, J. (2025). Scientific Reports, 15, 41721. DOI: 10.1038/s41598-025-26910-4	ML recommenders suffer from inadequate feature selection and class imbalance	Build an interpretable system for independent fertilizer and crop recommendation	TabNet, SHAP, SMOTE, iterative imputation, 5-fold CV; Random Forest baseline	95.24% accuracy for fertilizer and 96.21% for crop recommendation, above conventional classifiers	Use live IoT sensor data and adaptive updates
2	Fertilizer and Crop Recommendation System using Machine Learning	Sucharitha, M., et al. (2025). In Soft Computing and Signal Processing (ICSCSP 2024), LNNS 1200, pp. 579–588. DOI: 10.1007/978-981-97-9926-8_44	Precise fertilizer management is needed to cut resource waste	Predict the best fertilizer from agronomic parameters	Random Forest on soil type, nutrients, weather and crop type; deployed in a web app	Authors report high accuracy and a usable web interface	Validate on wider regions and crops (suggested)
3	Incorporating soil information with machine learning for crop recommendation to improve agricultural output	Afzal, H., Amjad, M., Raza, A., et al. (2025). Scientific Reports, 15, 8560. DOI: 10.1038/s41598-025-88676-z	Crop choice must account for soil and environmental conditions	Recommend the best of 22 crops	RFXG ensemble of Random Forest and XGBoost on temperature, humidity and pH	Ensemble learning is effective for multi-class crop recommendation	Extend to fertilizer recommendation (suggested)
4	Can machine learning models	Tanaka, T. S. T., & Heuvelink, G. B. M. (2024). Precision	Site-specific input rates need	Test how algorithm and	Random Forest,	Compares algorithm and	Test on more crops and sites

	provide accurate fertilizer recommendations ?	Agriculture, 25(4), 1839–1856. DOI: 10.1007/s11119-024-10136-x	accurate yield-response models	covariate choice affect yield and optimal input-rate prediction	XGBoost, SVR and ANN with 255 covariate sets from 8 soil properties; real winter-wheat on-farm data	covariate effects on real on-farm fertilizer data	(suggested)
5	Enhancing crop recommendation systems with explainable artificial intelligence: a study on agricultural decision-making	Shams, M. Y., Gamel, S. A., & Talaat, F. M. (2024). Neural Computing and Applications, 36(11), 5695–5714. DOI: 10.1007/s00521-023-09391-2	Current crop recommenders lack transparency and interpretability	Develop XAI-CROP, an explainable recommendation algorithm	Explainable AI with tree-based ML models	Reported better accuracy and transparency than other systems	Apply explainability to fertilizer choice (suggested)
6	Enhanced deep learning based precision agriculture: a decision support system for enhancing crop recommendation accuracy using convolutional neural networks (CNN)	Amin, M. N., Memon, S., Ali, A., Khan, H., Joshi, R., Ran, M. T. A., & Alsaawy, Y. (2024). Journal of Mechanics of Continua and Mathematical Sciences, 19(9), 166–187. DOI: 10.26782/jmcms.2024.09.00014	Farming decisions need accurate, interpretable crop recommendation	Improve crop-recommendation accuracy with a CNN and XAI principles	CNN on a Kaggle dataset of N, P, K, temperature, humidity, pH and rainfall	CNN improves crop-recommendation precision	Compare with tree-based models (suggested)
7	Machine learning based recommendation of agricultural and horticultural crop farming in India under the regime of NPK, soil pH and three climatic variables	Dey, B., Ferdous, J., & Ahmed, R. (2024). Heliyon, 10(3), e25112. DOI: verify on the publisher page	Recommending crops or missing nutrients for a given site	Compare five ML models for crop selection	SVM, XGBoost, Random Forest, KNN and Decision Tree on 11 agricultural and 10 horticultural crops	XGBoost precision of 99.09%, 99.3% and 98.51% (agricultural, horticultural, combined); evaluate crop categories separately	Build a user-friendly cloud-based AI interface
8	Crop recommendation with machine learning: leveraging environmental and economic factors for optimal crop selection	Sam, S., & D'Abreo, S. M. (2025). arXiv:2505.21201 (preprint)	Existing recommenders focus on environmental factors and narrow geography	Add economic factors for 19 crops across 15 Indian states	Random Forest and SVM with 10-fold CV, time-series split and lag variables	Random Forest with lag variables is the preferred model	Test with more regions and price data (suggested)
9	Comparative analysis of machine learning approaches for crop recommendation in sustainable agriculture in India	Yadav, S., Singh, H. K., Garg, A., & Khan, M. M. (2026). Proceedings of ISSF 2026. DOI: 10.2991/978-94-6239-693-7_57	Farmers need data-driven crop selection from soil and weather data	Compare many classifiers on N, P, K, pH, temperature, humidity and rainfall	Decision Tree, Random Forest, Gradient Boosting, Extra Trees, AdaBoost and XGBoost	ML methods can support data-driven crop selection and sustainable farming	Add real-time sensing (suggested)

10	Machine learning-based approaches to enhance the soil fertility — a review	Sujatha, M., & Jaidhar, C. D. (2024). Expert Systems with Applications, 240, 122557. DOI: verify on the publisher page	Dependence on costly lab tests and few low-cost, high-accuracy models	Review ML and DL for soil-fertility classification and fertilizer prediction	Literature review of ML and DL approaches	ML and DL can classify soil fertility accurately to guide fertilizer use	Build low-cost, high-accuracy classification models

3. Methodology

3.1 Proposed System Flow Diagram

The implemented workflow is:

Dataset → Data inspection and missing-value check → EDA → Label encoding → Feature scaling → Feature/targetseparation → Stratified 80:20 split → Training of Logistic Regression, Decision Tree and Random Forest → Test-set evaluation → Model comparison → Best-model analysis (confusion matrix, feature importance) → Fertilizer recommendation function

3.2 Dataset Description

The dataset has 1,000 records and 9 columns: 8 input features and the target Fertilizer_Name. It is generated in code with a fixed random seed (42) so the project runs without any file upload. The generator draws each feature uniformly from the ranges below. To use the real Kaggle fertilizer-prediction CSV instead, the generation block can be replaced by a single file-loading line.

Crop_Type	Categorical	Rice, Wheat, Sugarcane, Cotton, Maize, Millets, Pulses
Feature	**Type**	**Values / range**
Fertilizer_Name (target)	Categorical	10-26-26, 14-35-14, 17-17-17, 20-20, 28-28, DAP, Urea
Humidity	Numerical	30–89
Moisture	Numerical	20–69
Nitrogen	Numerical	0–44
Phosphorous	Numerical	0–44
Potassium	Numerical	0–44
Soil_Type	Categorical	Sandy, Loamy, Black, Red, Clayey
Temperature	Numerical	20–44

The fertilizer label is assigned by rules on the nitrogen, phosphorous and potassium levels (for example, low N, P and K gives 10-26-26, and high N with low P and K gives Urea). 5% of the labels (50 records) are then replaced with random fertilizer names so the task is not trivially perfect. The class counts are 28-28 (338), DAP (207), 17-17-17 (186), 14-35-14 (116), 20-20 (75), Urea (41) and 10-26-26 (37).

3.2 Experimental Setup

The project is implemented in Python as a single notebook script in Google Colab, a cloud-based environment that runs in the browser and needs no local installation. The code is organised into ten steps: importing libraries, building the dataset, exploratory analysis, preprocessing, train-test splitting, model training, model comparison, confusion-matrix analysis, feature importance, and a live prediction demo.

NumPy and pandas are used for data generation and handling, Matplotlib and seaborn for plots, and scikit-learn for label encoding, standardisation, the train-test split, the three classifiers and the evaluation metrics. A fixed random seed (42) is used for data generation, the train-test split and every model, so the results can be reproduced exactly on a rerun. The exact library versions and the hardware used were not recorded in the project files, so they are not stated here.

3.4 Data Preprocessing

- Data inspection covered shape, column names, data types, a statistical summary and a missing-value count.
- Missing values: 0 in every column, so no imputation was required. No duplicate rows were found (0).
- No separate outlier treatment is applied; all features are bounded by construction.
- Categorical encoding: Soil_Type, Crop_Type and the target are converted with LabelEncoder (5, 7 and 7 classes respectively).
- Feature scaling: StandardScaler is applied to all 8 input features, giving each zero mean and unit variance. Scaling is applied to the full feature matrix before the split.
- Split: 80:20 train-test split with random_state=42 and stratification on the target, giving 800 training and 200 test records.

3.5 Exploratory Data Analysis

The EDA step prints the data info, missing-value counts and statistical summary, plots the fertilizer class distribution as a count plot, and plots a correlation heatmap of the six numerical features. The class distribution is imbalanced, with 28-28 the largest class and Urea and 10-26-26 the smallest. The correlations between the numerical features are all close to zero (the largest magnitude is 0.09, between Temperature and Moisture), which is expected because each feature is generated independently.

3.6 Feature Engineering and Feature Selection (if applicable)

No derived features or automated feature-selection method are used. All 8 input features are retained. Scaling and label encoding are the only transformations.

3.7 Model Building and Training

Three classifiers are trained on the same training data and tested on the same test set:

Model	Configuration
Logistic Regression	max_iter=1000, random_state=42
Decision Tree	max_depth=8, random_state=42
Random Forest	n_estimators=200, max_depth=10, random_state=42

3.8 Hyperparameter Tuning(if applicable)

No GridSearchCV or RandomizedSearchCV is used. The models run with fixed parameters (above), with tree depth limited to reduce overfitting. No separately optimised hyperparameter set is reported.

3.9 Model Evaluation

Each model is evaluated on the 200 held-out test records with accuracy and weighted-average precision, recall and F1-score (weighted averages account for the unequal class sizes). The best model by accuracy is selected automatically. For

that model the project also produces a confusion matrix, a per-class classification report and a feature-importance chart. ROC-AUC is not computed.

4. Results and Discussion

4.1 Preprocessing Results

The dataset has 1,000 rows and 9 columns, with 0 missing values and 0 duplicate rows. Label encoding mapped the five soil types to 0–4 (Black, Clayey, Loamy, Red, Sandy), the seven crops to 0–6 (Cotton, Maize, Millets, Pulses, Rice, Sugarcane, Wheat) and the seven fertilizers to 0–6 (10-26-26, 14-35-14, 17-17-17, 20-20, 28-28, DAP, Urea). After standardisation, the stratified split produced 800 training and 200 test records.

4.2 Model Performance

Random Forest is the best model, with 94.5% accuracy and a weighted F1-score of 0.941 on the 200 test records. Precision, recall and F1-score are weighted averages over the seven classes.

Model	Accuracy	Precision	Recall	F1-score
Random Forest	94.5%	0.947	0.945	0.941
Decision Tree	91.0%	0.911	0.910	0.909
Logistic Regression	71.0%	0.695	0.710	0.696

The per-class report of the Random Forest shows where it is strong and weak:

Fertilizer	Precision	Recall	F1-score	Test samples
10-26-26	0.80	0.57	0.67	7
14-35-14	1.00	0.96	0.98	23
17-17-17	0.95	1.00	0.97	37
20-20	1.00	0.87	0.93	15
28-28	0.91	1.00	0.95	68
DAP	0.98	0.98	0.98	42
Urea	1.00	0.50	0.67	8

Macro-averaged recall (0.84) and F1 (0.88) are lower than the weighted values because the two smallest classes, Urea and 10-26-26, are often predicted as larger classes.

4.3 Visual Results and discussion

The script generates the class-distribution count plot, the correlation heatmap, the accuracy comparison bar chart of the three models, the Random Forest confusion matrix and the feature-importance chart. Insert these figures from the Colab output here.

The feature-importance values of the Random Forest show that the nutrient levels drive the recommendation:

Feature	Importance
Nitrogen	0.281
Potassium	0.279
Phosphorous	0.210
Humidity	0.060
Moisture	0.057
Temperature	0.050
Crop_Type	0.034
Soil_Type	0.028

Nitrogen, potassium and phosphorous together account for 77% of the total importance. The tree-based models beat Logistic Regression by 20 to 24 percentage points of accuracy, which indicates that the fertilizer choice depends on threshold-style, non-linear combinations of nutrient levels that a linear model cannot capture.

The high accuracy should be read with care. The labels are generated from rules on N, P and K, and 5% are randomly changed, so the dataset favours tree-based models and the results do not show performance on real farm data. Validation on the real Kaggle dataset or field data is needed before practical use.

5. Conclusion

The project implements a multi-class fertilizer recommendation pipeline using eight soil, crop and weather features, and Random Forest is the best of the three classifiers at 94.5% test accuracy. The data is inspected, categorical features are label-encoded, numerical features are standardised, and a stratified 80:20 split is used for training and testing. Logistic Regression, Decision Tree and Random Forest are compared on accuracy, precision, recall and F1-score; Random Forest scores 94.5% accuracy and 0.941 weighted F1, the Decision Tree 91.0% and 0.909, and Logistic Regression 71.0% and 0.696.

Nitrogen, potassium and phosphorous are the most important features, and the smallest classes (Urea and 10-26-26) are the hardest to recognise. The best model is wrapped in a recommendation function that turns raw soil and crop inputs into a fertilizer suggestion. Because the dataset is rule-based and synthetic, the results demonstrate the method rather than field performance, and the output should support, not replace, expert and laboratory advice.

6. Future Work

- Train and validate on the real Kaggle fertilizer-prediction dataset and on field data from different regions and seasons.
- Fit the StandardScaler on the training data only (for example, with a scikit-learn Pipeline) to prevent data leakage, and add cross-validation for more stable estimates.
- Perform systematic hyperparameter tuning with GridSearchCV or RandomizedSearchCV.
- Handle class imbalance (class weights or SMOTE) to improve recall for Urea and 10-26-26.
- Compare additional models such as SVM, KNN, Gradient Boosting and XGBoost, and add ROC-AUC.
- Add explainable-AI methods such as SHAP or LIME to explain each recommendation.
- Recommend fertilizer quantity, not only the name, and build a web or mobile interface with regional-language support for farmers.

7.References

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